

DBMS Semester Examination 2025

Database Management Systems (DBMS) · Semester Examination · 2025

Max. Marks	75	Time	3 hours
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Answer ALL questions from Part A and any FIVE from Part B.

Part A (25 marks)

1. Illustrate Database System Architecture with a worked example. (5 marks)
2. Write short notes on Relational Algebra. (5 marks)
3. What is meant by Functional Dependencies? Discuss its advantages and limitations. (5 marks)
4. Illustrate ACID Properties with a worked example. (5 marks)
5. What is meant by File Organization? Discuss its advantages and limitations. (5 marks)

Part B

1. Write short notes on Database System Architecture. (Unit 1)
2. Write short notes on Data Independence. (Unit 1)
3. Explain Relational Algebra with a neat diagram and a suitable example. (Unit 2)
4. What is meant by Joins? Discuss its advantages and limitations. (Unit 2)
5. What is meant by Functional Dependencies? Discuss its advantages and limitations. (Unit 3)
6. Discuss the role of Closure of Attributes in Normalization & Schema Refinement. (Unit 3)
7. Illustrate ACID Properties with a worked example. (Unit 4)
8. Discuss the role of Serializability in Transactions & Concurrency Control. (Unit 4)
9. What is meant by File Organization? Discuss its advantages and limitations. (Unit 5)
10. Compare B+ Tree Indexes with File Organization. When would you prefer one over the other? (Unit 5)